AWS Project: Host Your First Static Website

Lab Guide for Cloud Beginners

This lab is designed for individuals new to cloud computing. By the end of this session, you will have a live, accessible website hosted on the global AWS infrastructure using Amazon S3.

Lab Objective

Create a simple multi-page website using HTML and CSS, then deploy it to the cloud using:

- Amazon S3 (Simple Storage Service)

No backend development or programming knowledge is required.

Prerequisites

Requirement	Specification
AWS Account	AWS Free Tier Account
Hardware	A Windows PC
Connectivity	Stable Internet Connection
Software	Any Web Browser (Chrome/Edge)

Phase 1: Local Environment Setup

Step 1: Create a Project Folder

On your Desktop:

1. Right-click on an empty area.
2. Click **New** → **Folder**.
3. Name the folder: `MyWebsite`

Step 2: Create index.html

1. Open the `MyWebsite` folder.
2. Right-click and select **New** → **Text Document**.
3. Rename it to `index.html`. (Note: If you see `index.html.txt`, ensure "File name extensions" is enabled in Windows Explorer View settings and remove the `.txt`).
4. Open with Notepad and paste the following:

```
None
<!DOCTYPE html>
<html>
<head>
  <title>My First AWS Website</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Welcome to My Website</h1>
  <h2>Hosted on Amazon S3</h2>
  <p>This is my first AWS Static Website.</p>
  <a href="about.html">About</a> | <a href="contact.html">Contact</a>
</body>
</html>
```

Step 3: Create about.html

Create another text document named `about.html` and paste:

```
None
<!DOCTYPE html>
<html>
<head>
  <title>About</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>About Me</h1>
  <p>I am learning AWS Cloud.</p>
  <a href="index.html">Home</a>
</body>
</html>
```

Step 4: Create contact.html

Create a document named `contact.html` and paste:

```
None
<!DOCTYPE html>
<html>
<head>
  <title>Contact</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Contact</h1>
  <p>Email: <span type="placeholder"
placeholder-type="person"></span>@gmail.com</p>
  <p>Phone: 9876543210</p>
  <a href="index.html">Home</a>
</body>
</html>
```

Step 5: Create style.css

Create a document named `style.css` to add visual styling:

```
None
body {
  font-family: Arial;
  background: #f2f2f2;
  text-align: center;
  margin-top: 50px;
}
h1 { color: blue; }
h2 { color: green; }
p { font-size: 20px; }
a {
  text-decoration: none;
  font-size: 20px;
  margin: 10px;
}
```

Step 6: Verify Structure & Test Locally

Your folder should contain: `index.html`, `about.html`, `contact.html`, and `style.css`. Double-click `index.html` to ensure the links and styles work in your browser before moving to AWS.

Phase 2: AWS Deployment

Step 7: Create an S3 Bucket

1. Sign in to the AWS Management Console.
2. Search for **S3** and click **Create bucket**.
3. **Bucket name:** `my-first-website-` followed by your name (e.g., `my-first-website-pravin`).
4. **Region:** Asia Pacific (Mumbai) `ap-south-1`.
5. Leave other settings as default and click **Create bucket**.

Step 8: Upload and Configure

1. Open your new bucket and click **Upload**.
2. Add all four files from your `MyWebsite` folder and click **Upload**.
3. Go to the **Properties** tab, scroll to **Static website hosting**, and click **Edit**.

4. Select **Enable**.
5. Set Index document to `index.html` and click **Save changes**.

Step 9: Permissions & Public Access

1. Go to the **Permissions** tab.
2. Click **Edit** under **Block public access**. Uncheck **Block all public access**, acknowledge the warning, and click **Save**.
3. Scroll down to **Bucket Policy**, click **Edit**, and paste:

```
None
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicRead",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
    }
  ]
}
```

Note: Replace `YOUR-BUCKET-NAME` with your actual bucket name.

Step 10: Launch

Go back to **Properties** → **Static website hosting**. Copy the **Bucket website endpoint** and open it in a new tab. Your website is now live!

Learning Outcomes

By completing this project on 🗓 Date, you have mastered:

- Standard web file organization.
- S3 bucket creation and object management.
- Configuring IAM-based bucket policies for public read access.

- Deploying serverless static content at scale.